

LESSON 5.10

SOLVING PROBLEMS INVOLVING REAL-WORLD SCALE DRAWINGS



LESSON INTRODUCTION

MA.7.GR.1.5 Solve mathematical and real-world problems involving dimensions and areas of geometric figures, including scale drawings and scale factors.

LEARNING TARGETS

- I can solve real-world problems involving scale drawings.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.8.GR.2.4 Solve mathematical and real-world problems involving proportional relationships between similar triangles.

ESSENTIAL QUESTIONS

- How can you determine if a model has an accurate scale?
- Why would exactness of measurements using scale factor be necessary in the real world?

LESSON NARRATIVE

In this lesson, students solve real-world problems involving scale drawings. They first explore famous landmarks drawn using a common scale. They use the scale to determine approximations of the actual heights of the landmarks. Students then use what they have learned about scale models to determine whether a model is actually created to scale before practicing both collaboratively and independently to solve problems applying these concepts.

COMPONENT SUMMARY

5.10.1 WARM-UP (~5 MIN.)

Career Profile: Architectural Designer

5.10.3 GUIDED PRACTICE (10-15 MIN.)

Connection of scale factor to toy models and house blueprints

5.10.5 WRAP-UP (~5 MIN.)

Write a short summary on scale factors and drawings

5.10.2 GUIDED INSTRUCTION (10-15 MIN.)

Use scale factor with real-world landmarks

5.10.4 YOUR TURN! (10-15 MIN.)

Complete problems to determine actual dimensions of real-world objects and places

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Career Profile video
- Measuring tools for 5.10.2 Guided Instruction

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not grasp the idea that most architectural pieces have more than three sides or more than two dimensions.

THINGS TO CONSIDER

- It may be useful to note the differences between scale drawings, perspective drawings, and photographs. Some photographs can be taken at different perspectives and can make things appear shorter or taller than they really are.
- When pictures are taken at an angle and not just straight on, it is difficult to determine the actual height.
- When there are no other reference points such as trees or people in photographs, it can also make it difficult to determine the dimensions from a photograph or perspective drawing.

LITERACY AND LANGUAGE ROUTINES

Take Turns

5.10.2 Guided Instruction provides an opportunity for students to engage in a Take Turns routine. Students have the opportunity to engage in the work of the section while justifying and explaining how they arrive at each answer. This deepens understanding of the content.

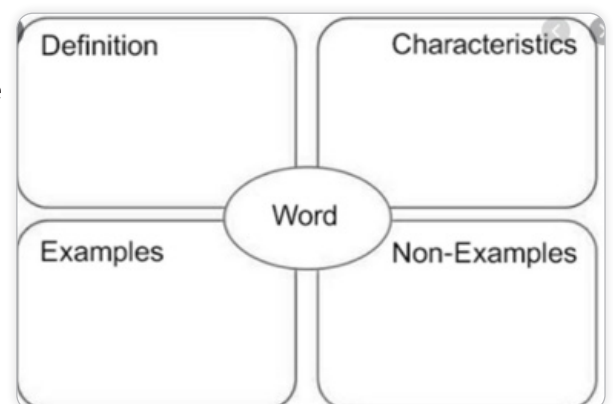
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

The entire lesson is created around the idea that scale factor can be used when creating scale drawings and models of real-world landmarks. From the 5.10.2 Guided Instruction throughout the rest of the lesson, students are offered multiple opportunities to see and use these concepts in real-world contexts.

DIFFERENTIATE INSTRUCTION

The 5.10.5 Wrap-Up encourages students to write a Two-Dollar Summary explaining scale drawings. For students who may struggle with written expression, consider alternative means of completing this Cool Down. Some students may choose to record their summary using an app, like FlipGrid, or pose it orally to the teacher. Some students may find it helpful to organize their thoughts on a graphic organizer like a Frayer Model, with which they can provide examples and nonexamples and list characteristics.



5.10.1 WARM-UP: A DAY IN THE LIFE

TEACHER MOVES

This could create an opportunity to discuss connections math not only has to other subjects but also to the world in general around the students. The idea of math being something that is considered beautiful may be a new idea that students could also explore.



5.10.1 Warm-Up: A Day in the Life

Rachele Louis is an architectural designer who specializes in designing the cores and shells of buildings, or as she calls it, the “skin and bones.” Geometry is key to her work, especially in the initial design phases and when she’s trying to connect forms. She states that there is an art to architecture as well. The appearance of a building defines people’s understanding of the space and evokes certain feelings.

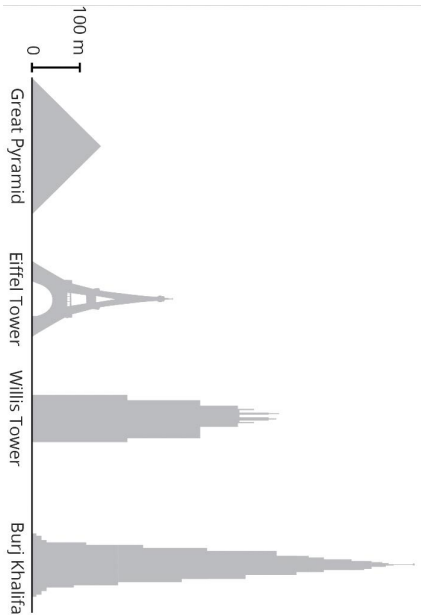
Rachele shares that this integration of art, geometry, and design is her favorite part of the job.

1. Can you think of any examples of how art is used in architecture?
Answers will vary. Yes, there are many examples of art in architecture, there are many patterns that are used – such as in tiles or in cut outs. There are also stylistic differences, such as Spanish tile roofing.
2. What geometric shapes do you see in your classroom?
Answers will vary. I see rectangles in the ceiling. I see squares on the tile on the ground. I see circles in the light fixtures.



5.10.2 Guided Instruction: Scale in Architecture

Shown is a scale drawing of some of the world’s tallest structures.



1. Turn your paper sideways and measure the line segment that shows the scale with a ruler to the nearest tenth of a centimeter (cm).
The line segment is approximately 1.1cm.

2. Express the scale of the drawing using numbers and words.
The scale is 100m to 1.1cm or $\frac{100m}{1.1cm}$.
3. Complete the table using the scaled drawing. Include units.

	Actual Height
Great Pyramid	$1.5cm \times \frac{100m}{1.1cm} \approx 136m$
Eiffel Tower	$3.3cm \times \frac{100m}{1.1cm} \approx 300m$
Willis Tower	$5.7cm \times \frac{100m}{1.1cm} \approx 518m$
Burj Khalifa	$8.9cm \times \frac{100m}{1.1cm} \approx 809m$

4. About how much taller is the Burj Khalifa than the Eiffel Tower?
The Burj Khalifa is approximately 509 meters taller than the Eiffel Tower.
5. The tallest mountain in the United States, Mount Denali in Alaska, is about 6,190 meters (m) tall. If this mountain were shown on the scale drawing, explain how its height would compare to the heights of the other structures.
To find the height of Mount Denali on the scale drawing, multiply the actual height by the reciprocal of the scale factor: $6,190m \times \frac{1.1cm}{100m} \approx 68.1cm$. Comparing 68.1cm and 8.9cm, we can say that if Mount Denali were shown on the scale drawing, it would be about 8 times taller than the scale drawing of the Burj Khalifa.

5.10.2 GUIDED INSTRUCTION: SCALE IN ARCHITECTURE

TEACHER MOVES

Students will need rulers with centimeter and millimeter markings on them.

TEACHER MOVES

Take Turns

Consider question 3 an opportunity to engage the class in a Take Turns routine. Partner students together and have them take turns finding the actual heights of each building. Encourage students to discuss their strategies for solving and to correct any errors in their partner’s work. This will deepen understanding and encourage mathematical discourse.

5.10.3 GUIDED PRACTICE: ACCURACY OF MODELS

QUESTIONING

For students who finish early, consider giving the actual dimensions of the Statue of Liberty to calculate the dimensions of a scaled replica.



5.10.3 Guided Practice: Accuracy of Models

1. Lego has created a line of products that are scales of important architectural pieces from around the world. Myra saw the Lego statue shown and wondered if it was a scale model of the actual Statue of Liberty.



Lego Statue
height 17 inches (in.)
width 5 in.
depth 5 in.



Actual Statue
height 305 feet (ft)
width 35 ft
depth 35 ft

- a. Using the measurements given, determine if the Lego is a scale model of the actual Statue of Liberty.

The Lego is not a scale model of the actual Statue of Liberty. When the height is multiplied by the same scale factor (84in or 7ft), you get an answer of 1,428 inches, or 119 feet, rather than the height required to reach the actual statue which would be 3,660 inches or 305 feet.

- b. Explain whether the measurements are enough to determine exactness for the whole scale model.

These measurements are enough to determine whether the Lego is a scale model.

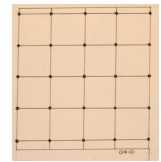
ENRICHMENT

Solidify student understanding through questioning. What shapes do the grids of columns make in the Villa Savoye? (Squares.) Why would an architect need to ensure that a scale drawing of a house or building is correct?

2. Villa Savoye is a house designed by the Swiss architect Le Corbusier. It was built between 1928 and 1931, and is considered a prime example of modern architecture and the International Style.



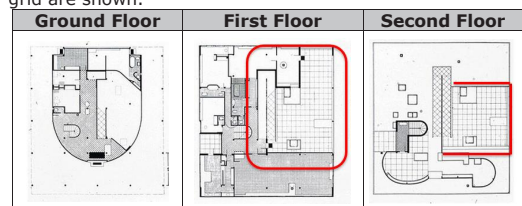
The columns of the structure were placed in an arrangement the architect called a "square grid." An image of the grid is shown.

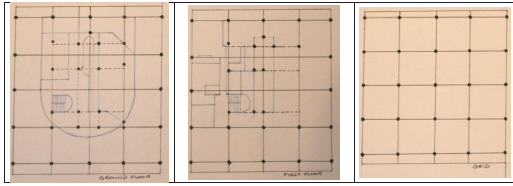


- a. If the columns are 4.75 m apart, what is the approximate length and width of the actual house?

The columns create a square grid. When multiplied each side would have a length of approximately 19 m. If we wanted to round this to account for the overhang on the edges, we could round to a length and width of 20 m.

The floor plans of Villa Savoye and images of the square grid are shown.





- b. Approximately how many square meters (m^2) of concrete does the foundation cover?
The foundation of the house appears to take up approximately 16 squares. Each square has an area of about $22.6 m^2$, or $361 m^2$ in total.
- c. What is the approximate area of the living room (the tiled area inside the red boxed area on the first floor)?
The living room tiles of the first floor appear to take up approximately 12 squares. Each square has an area of about $22.6 m^2$, or $271 m^2$ in total.
- d. The plumber needs to put in all new plumbing in one room on the second floor. She only needs to lay the plumbing along three of the walls (the ones lined with red). Calculate approximately how many meters of pipe the plumber needs.
The walls identified take up approximately 5.5 side lengths (1.5 top, 2 side, 2 bottom), or 5.5×4.75 meters, which equals about $26.125 m$ of pipe.

5.10.3 GUIDED PRACTICE: ACCURACY OF MODELS (CONTINUED)

DIFFERENTIATE INSTRUCTION

As an Extension, consider having students create and share a scale drawing of their dream home by doing the following:

- Determine the dimensions the rooms of their dream home should have.
- Instruct them that the home must fit on an 8.5" x 11" piece of grid paper.
- Have the students determine what the scale should be so that the drawing will fit on the grid paper.
- Have them create the drawing of their home.
- The above can be extended further so that students create a three-dimensional replica.

5.10.4 YOUR TURN!

TEACHER MOVES

Provide measuring tools as needed for students to complete the first question. Students could also use a separate paper and mark each length on it to use in approximating different lengths from the drawings.

ENRICHMENT

If students are struggling to approximate the actual lengths, consider asking questions to guide their thinking. What scale factors are given? How can you use the scales given to approximate these measures?

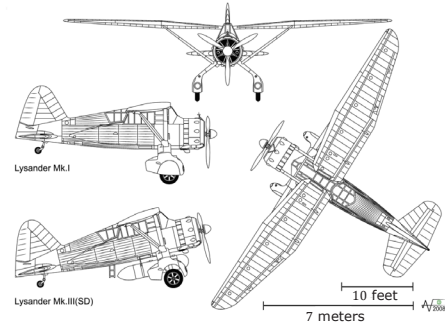
DIFFERENTIATE INSTRUCTION

Encourage the use of models or diagrams (such as a tape diagram) to represent images.



5.10.4 Your Turn!

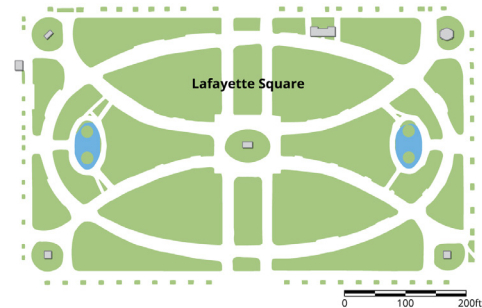
- The Westland Lysander was an aircraft used by the Royal Air Force in the 1930s. Here are some scale drawings that show the top, side, and front views of the Lysander.



Use the scales and scale drawings to approximate the actual lengths.

wingspan of the plane to the nearest foot	47 ft
height of the plane to the nearest foot	12 ft
length of the plane to the nearest meter	9 m

- Below is a scale map of Lafayette Square, a rectangular garden north of the White House.



- The scale is shown in the lower right corner. What are the actual side lengths of Lafayette Square in feet?

$$L \approx 14.8\text{cm} \times \frac{200\text{ft}}{4\text{cm}} \approx 740\text{ft}$$

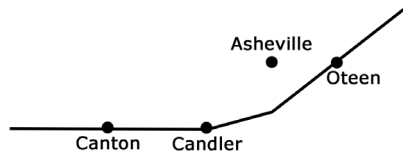
$$W \approx 9\text{cm} \times \frac{200\text{ft}}{4\text{cm}} \approx 450\text{ft}$$

- Use an inch ruler to measure the line segment of the graphic scale. About how many feet does one inch represent on this map?

Using the scale, we know $1\frac{1}{8}\text{in} = 200\text{ft}$.

So, $1\text{in} \approx 178\text{ft}$.

3. On the map below, $\frac{1}{4}$ in. represents one mile (mi). Candler, Canton, and Oteen are three cities on the map.



- a. If the distance between the actual towns of Candler and Canton is 9 mi., how far apart are Candler and Canton on the map?
If Candler and Canton are 9 miles apart, that would mean $9 \times \frac{1}{4}$ inches or $2\frac{1}{4}$ inches apart.
- b. If Candler and Oteen are $3\frac{1}{2}$ in. apart on the map, what is the actual distance between Candler and Oteen in miles?
Candler and Oteen are $3\frac{1}{2}$ inches apart, which is 14 segments of $\frac{1}{4}$ in or 14 miles according to the scale.

5.10.4 YOUR TURN! (CONTINUED)

TEACHER MOVES

For most of this lesson, students have been finding the measurements of the actual objects or places. Question 3 requires students to find the lengths in the scaled drawing based off the distance between the actual towns, reversing the pattern students may be used to. Students may make the mistake of assuming they are still finding the actual distance (vs. the distance on the map). Observe whether any students make this error and ask scaffolding questions if needed.



5.10.5 Wrap-Up: Cool Down

1. Write a two-dollar (or more) summary about scale drawings. Each word is worth 10 cents. Be sure to include how scale drawings are used, an example, and how to find measurements from a scale drawing.
Scale drawings are used to create smaller versions of actual objects. Dimensions of scale drawing can be multiplied by the same scale factor to get the dimensions of the actual object.

5.10.5 WRAP UP: COOL DOWN

DIFFERENTIATE INSTRUCTION

For students who may struggle with written expression, consider alternative means of completing this Cool Down. Some students may choose to record their summary using an app, like FlipGrid, or pose it orally to the teacher. Some students may find it helpful to organize their thoughts on a graphic organizer like a Frayer Model, with which they can provide examples and nonexamples and list characteristics.